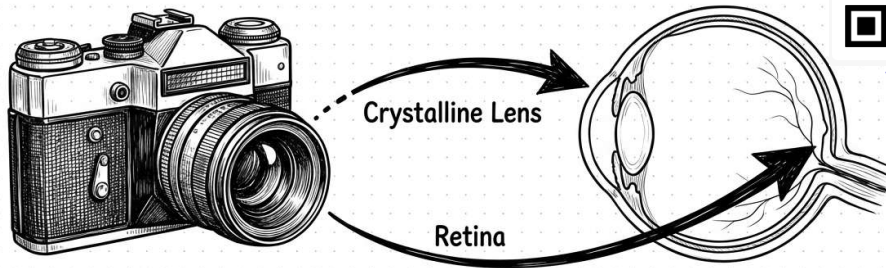


10.1 THE HUMAN EYE



website

Definition: The human eye is a natural optical instrument functioning like a camera, using a crystalline lens to form inverted, real images on a light-sensitive screen called the retina.

Real-life application: Enables us to identify colours, shapes, and distances; arguably our most significant sense organ.

Important Point: The eyeball is approximately spherical in shape with a diameter of about 2.3 cm.

Exam Alert: MCQs often ask for the nature of the image formed on the retina (Answer: Real and Inverted).

Anatomy of the Human Eye

Cornea:

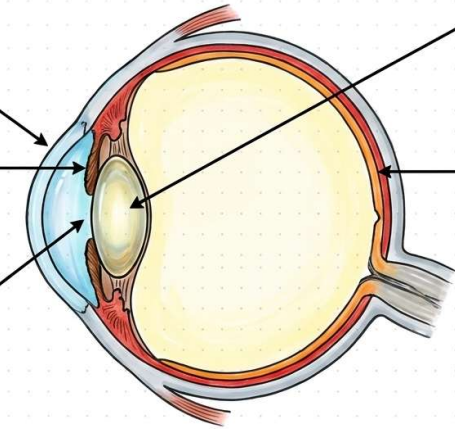
Transparent front bulge. Most refraction occurs at its outer surface.

Iris:

Dark muscular diaphragm. Controls the size of the pupil.

Pupil:

Window of the eye. Regulates the amount of light entering.



Crystalline Lens:

Fibrous, jelly-like. Provides fine focal adjustment.

Retina:

Delicate membrane screen with enormous numbers of light-sensitive cells.

Exam Alert: Do not confuse Iris and Pupil functions! Iris is the muscle that moves; Pupil is the opening that changes size.

Signal Processing in the Eye

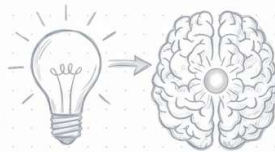
How does the image reach the brain?

Step 1: Illumination - Light falls on the retina.

Step 2: Activation - Light-sensitive cells get activated and generate electrical signals.

Step 3: Transmission - Signals are sent to the brain via the Optic Nerve.

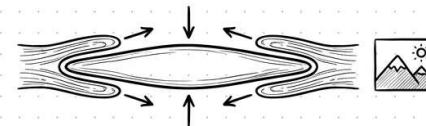
Step 4: Interpretation - The brain interprets these signals, processing the inverted image so we perceive objects as they are (upright).



10.1.1 Power of Accommodation

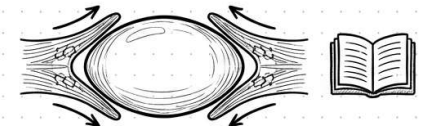
Definition: The ability of the eye lens to adjust its focal length to clearly see both near and distant objects.

Mechanism (The Role of Ciliary Muscles):



Distant Vision:

- Ciliary muscles relax
- Lens becomes thin
- Focal length increases



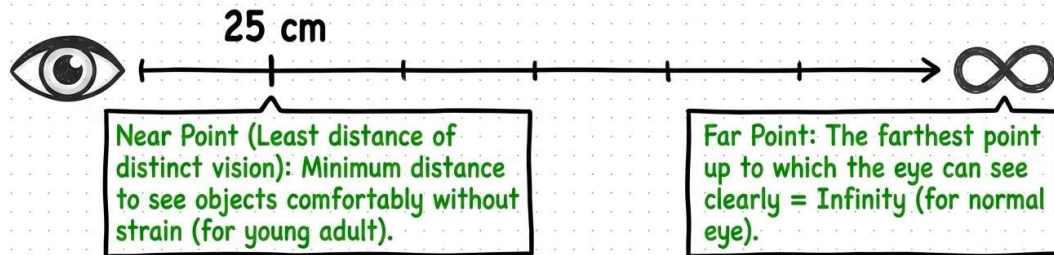
Near Vision:

- Ciliary muscles contract
- Lens becomes thick
- Focal length decreases

Memory Trick: Contract = Close objects.



Vision Limits & Cataract

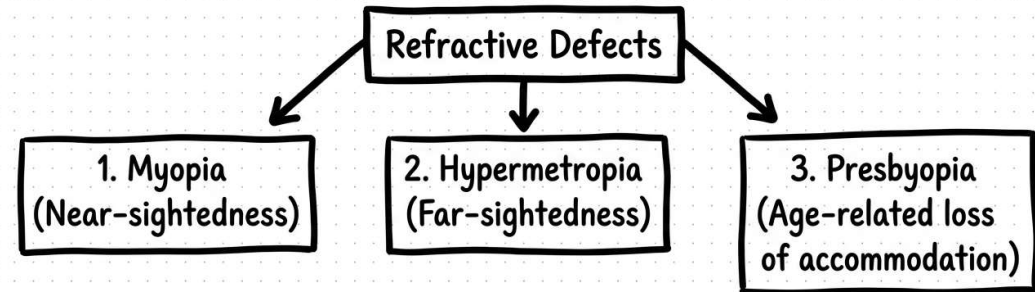


Ageing & Cataract:

- What is it? Crystalline lens becomes milky and cloudy in old age.
- Result: Partial or complete loss of vision.
- Correction: Possible to restore vision through cataract surgery.

10.2 DEFECTS OF VISION AND THEIR CORRECTION

Why do they happen? The eye gradually loses its power of accommodation. Vision becomes blurred due to refractive defects.



Must Remember Line: All three are corrected using suitable spherical lenses!

(a) Myopia

Also known as: Near-sightedness



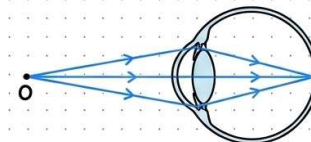
Condition: A person can see nearby objects clearly but cannot see distant objects distinctly. Far point is nearer than infinity.

Important Observation: The image of a distant object is formed **IN FRONT** of the retina, not on it.

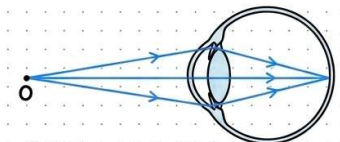
Causes (Board Exam Point):

1. Excessive curvature of the eye lens.
2. Elongation of the eyeball.

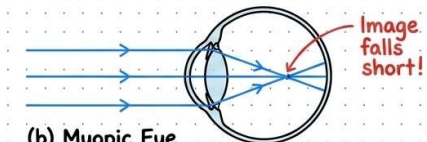
Myopia Ray Diagrams & Correction



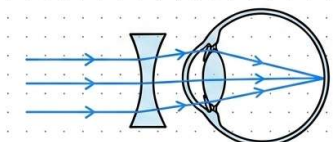
(a) Far point of a myopic eye



(a) Far point of a myopic eye



(b) Myopic Eye



(c) Correction for myopia

Correction: Corrected using a **CONCAVE LENS** of suitable power.

(b) Hypermetropia

Also known as: Far-sightedness



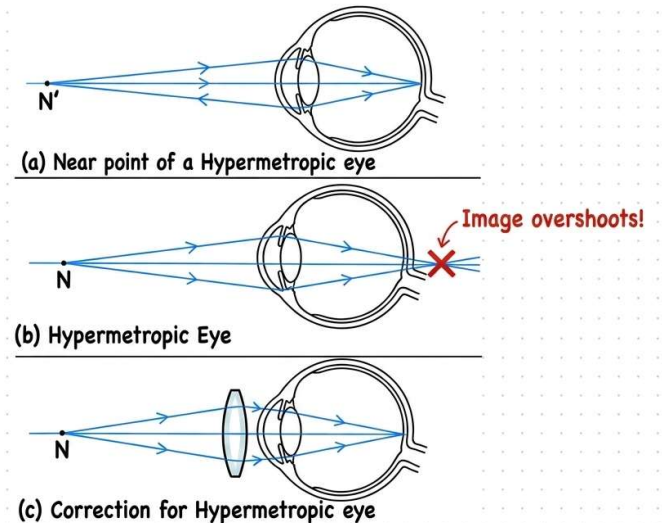
Condition: A person can see distant objects clearly but cannot see nearby objects distinctly. Near point is farther away than normal 25 cm.

- **Important Observation:** Light rays from a closeby object are focussed at a point **BEHIND** the retina.

Causes (Board Exam Point):

1. The focal length of the eye lens is too long.
2. The eyeball has become too small.

Hypermetropia Ray Diagrams & Correction



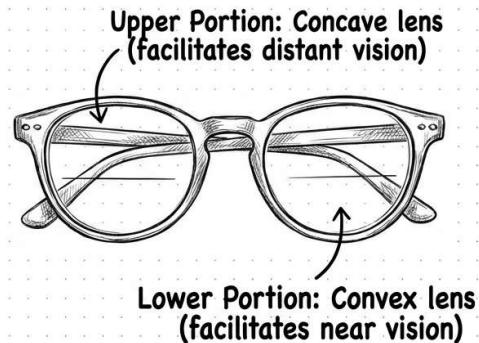
Correction: Corrected using a **CONVEX LENS** of appropriate power.

(c) Presbyopia

Condition: Decrease in power of accommodation with ageing. The near point gradually recedes away.

Causes: Weakening of the ciliary muscles and diminishing flexibility of the eye lens.

The "Double" Defect: Sometimes a person suffers from both myopia and hypermetropia.



Correction: Requires Bi-focal lenses.

Synthesis: The Ultimate Defect Comparison Matrix

Exam Tip: Memorize this table to solve 90% of defect-based MCQs!

Feature	Myopia	Hypermetropia	Presbyopia
Common Name	Near-sightedness	Far-sightedness	Old-age sightedness
Image Formation	In front of retina	Behind retina	Behind retina (usually)
Causes	Long eyeball / high lens curvature	Short eyeball / long focal length	Weak ciliary muscles / inflexible lens
Correction	Concave Lens	Convex Lens	Bi-focal Lenses

Eye Donation: Who CAN Donate?

Rules & Guidelines

- Any age group or sex.
- People who wear spectacles or had cataract surgery.
- Diabetics, hypertension, or asthma patients (without communicable diseases).

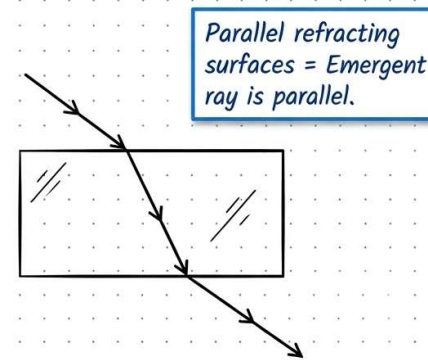
Who CANNOT Donate?

- Persons infected with AIDS, Hepatitis B/C, rabies, acute leukaemia, tetanus, cholera, meningitis, encephalitis.

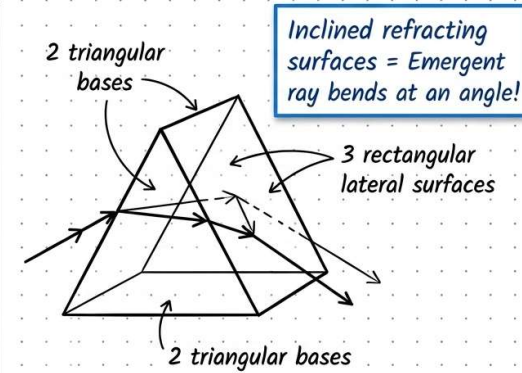
The Process:

- Eyes must be removed within 4-6 hours after death.
- Process takes only 10-15 minutes.
- Leaves NO disfigurement. Identities remain confidential.

10.3 REFRACTION OF LIGHT THROUGH A PRISM



Angle of the Prism ($\angle A$):
The angle between its two lateral faces.

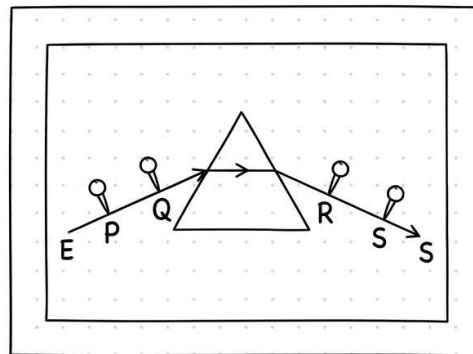


Board exams often ask for the key difference between slab and prism refraction. The key is the inclined surfaces of the prism!

ACTIVITY 10.1: Tracing the Prism Ray

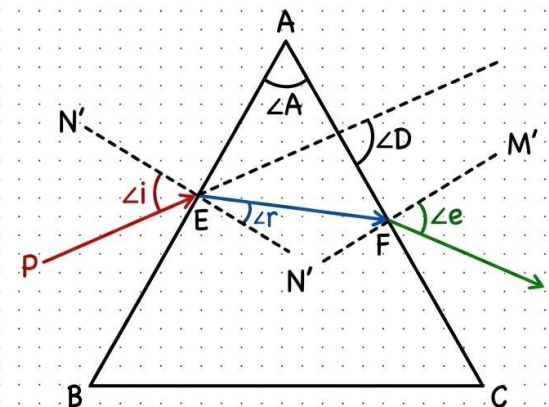
Aim: To study the refraction of light through a triangular glass prism.

1. Trace prism outline on white paper.
2. Draw straight line PE (incident ray) & fix pins P, Q.
3. Look through face AC to find images of P, Q.
4. Fix pins R, S in line with the images.
5. Remove prism, join points to create emergent ray FS.



Always trace the boundary of the prism first before removing it!

The Prism Ray Diagram (Anatomy of Refraction)



- $\angle i$ = Angle of incidence
- $\angle r$ = Angle of refraction
- $\angle e$ = Angle of emergence

Angle of Deviation ($\angle D$):

The peculiar shape of the prism makes the emergent ray bend at an angle to the direction of the incident ray. This angle is called the angle of deviation.

Mechanism:

- At surface AB: Ray bends towards normal (Rare to Dense).
- At surface AC: Ray bends away from normal (Dense to Rare).

10.4 DISPERSION OF WHITE LIGHT BY A GLASS PRISM

ACTIVITY 10.2: The Setup

Allow a narrow beam of sunlight (white light) through a cardboard slit to fall on a glass prism. Turn slowly until a band of colours appears on a screen.

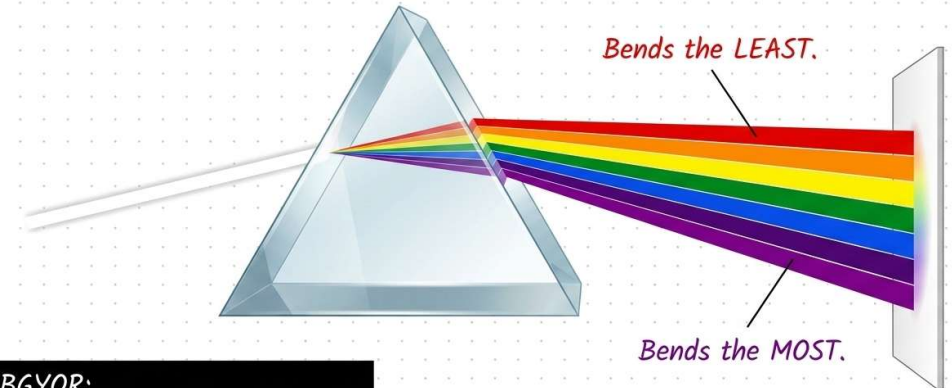
Dispersion: The splitting of light into its component colours.

Spectrum: The band of the coloured components of a light beam.

Mechanism: Why does this happen?

- Different colours of light bend through different angles with respect to the incident ray as they pass through a prism.

The VIBGYOR Spectrum & Bending Hierarchy



VIBGYOR:
Violet, Indigo, Blue, Green,
Yellow, Orange, Red
(Sequence from bottom to top).

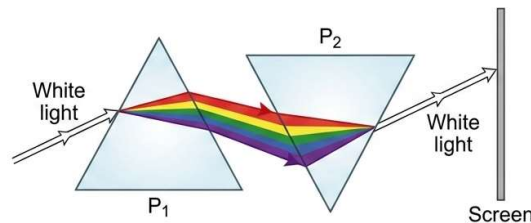
Thus the rays of each colour emerge along different paths and thus become distinct.

Isaac Newton's Recombination Experiment

Step 1: Newton used one prism to get a spectrum.

Step 2: Tried a second identical prism to split colors further. Result: Failed.

Step 3: Placed the second prism in an inverted position.

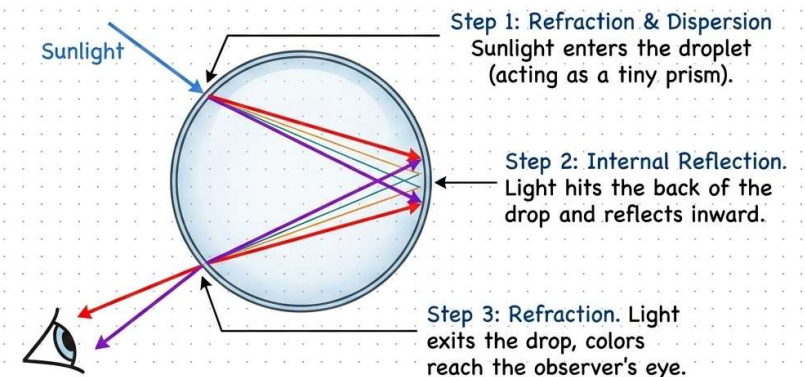


Discovery: Any light that gives a spectrum similar to sunlight is referred to as white light. The prism doesn't create colors; sunlight is made of seven colors!

Exam Alert: Students often forget to draw the second prism inverted. Always draw P1 upright and P2 upside down!

Natural Dispersion: Formation of a Rainbow

Prerequisite: A rainbow is ALWAYS formed in a direction opposite to that of the Sun.



Ensure you mention ALL THREE phenomena in your answer: Dispersion, Internal Reflection, and Refraction.

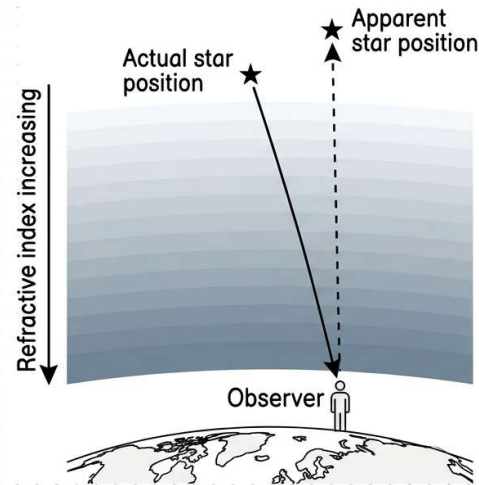
10.5 ATMOSPHERIC REFRACTION



- **The Cause:** The physical conditions of the refracting medium (air) are not stationary.
- **The Physics:** Hot air is lighter (less dense) = Lower refractive index. Cooler air above is denser = Higher refractive index.
- **The Effect:** Apparent random wavering or flickering of objects seen through this turbulent stream.

The **twinkling of stars** is this exact same phenomenon, just on a **massive, planetary scale!**

Why Do Stars Twinkle?



1. **Refraction:** Starlight enters a medium of gradually changing refractive index.
2. **Bending:** Atmosphere bends starlight towards the normal.
3. **Apparent Position:** The star appears slightly higher than its actual position.
4. **Fluctuation (Twinkling):** Because atmosphere is dynamic, the path of light varies. Star appears brighter, then fainter.

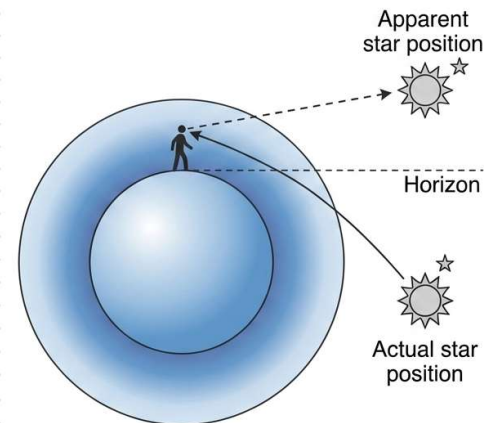
Key Assumption: Stars are very distant, so they approximate point-sized sources of light.

EXAMPLE / EXAM FAVORITE: Why don't planets twinkle?

Feature	Stars	Planets
Distance	Very distant	Much closer
Source Type	Point-sized sources	Extended sources (collection of many point-sized sources)
Visual Effect	Path fluctuates = Twinkling effect.	Total variation in light entering eye averages out to zero = Nullifies twinkling.

Must Remember: Always use the keyword “**Extended Source**” when explaining why planets do not twinkle.

Advance Sunrise and Delayed Sunset



- Sun is visible ~2 minutes before actual crossing of horizon.
- Sun is visible ~2 minutes after actual sunset.
- **Total Day Extension: 4 minutes!**

Bonus Fact: The apparent flattening of the Sun's disc at sunrise and sunset is caused by this exact same phenomenon (atmospheric refraction).



10.6 SCATTERING OF LIGHT & 10.6.1 Tyndall Effect



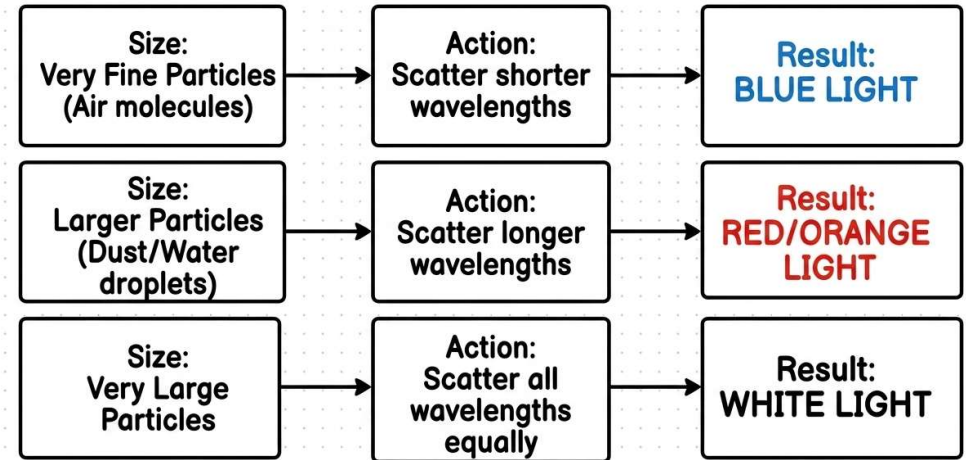
Tyndall Effect: The phenomenon of scattering of light by colloidal particles, making the path of the beam visible.

Real-Life Examples:

1. Fine beam of sunlight entering a smoke-filled room.
2. Sunlight passing through a canopy of a dense forest (mist scattering light).

Core Rule: The colour of the scattered light depends strictly on the size of the scattering particles.

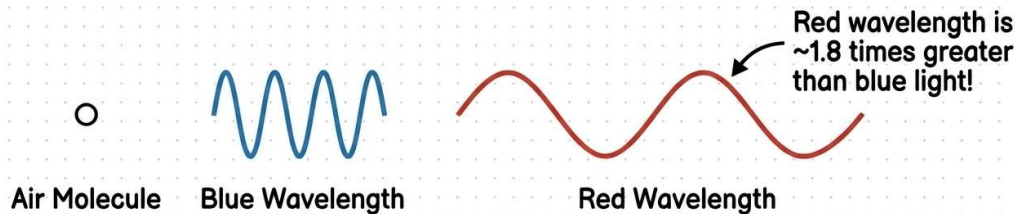
The Scattering Matrix: Particle Size vs. Wavelength



• for more resources -



10.6.2 Why is the colour of the clear Sky Blue?



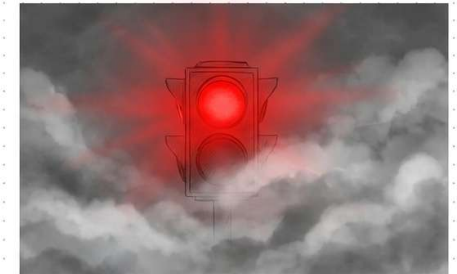
1. Molecules of air (Nitrogen/Oxygen) are smaller than the wavelength of visible light.
2. Sunlight passes through the atmosphere.
3. Fine particles strongly scatter the shorter wavelength (blue).
4. The scattered blue light enters our eyes.

Exam Alert: If Earth had no atmosphere, there would be **NO scattering**. The sky would look completely dark!

Scattering Exceptions: High Altitudes & Danger Signals



Sky appears dark at very high altitudes because scattering is not prominent (very few particles).



Why are danger signals red? Red is the least scattered by fog or smoke because of its long wavelength. It travels furthest and can be seen clearly from a distance.

Red = Longest Wavelength = Least Scattered